

ADC Peptide Mapper: A Workflow Guide

From FASTA to MRM Transition List

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1 Overview

ADC Peptide Mapper is an R Shiny application for in-silico peptide mapping of Antibody-Drug Conjugates (ADCs). Starting from a FASTA sequence, it performs enzymatic digestion, uniqueness checking against human / cynomolgus monkey / rat proteome backgrounds, Drug-to-Antibody Ratio (DAR) distribution modelling, and instrument-specific MRM/DIA transition list generation.

This vignette walks through the complete workflow for Trastuzumab-DXd using the built-in demo sequence.

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2 Installation

2.1 R Packages

```
install.packages(c(
  "shiny", "bs4Dash", "DT", "data.table", "openxlsx",
  "httr2", "stringr", "dplyr", "shinyjs", "htmltools"
))
```

2.2 Build Background Databases (one-time)

The uniqueness filter requires pre-digested reference proteomes. Build them once before launching the app. An internet connection is required; the download takes approximately 10 minutes.

```
setwd("path/to/ADC_Peptide_Mapper_v0.8")
source("build_background_db.R")
```

This creates three files in `data/`:

- `bg_human.rds` — Human (*Homo sapiens*), ~20,400 UniProt Swiss-Prot proteins
- `bg_monkey.rds` — *Cynomolgus* monkey (*Macaca fascicularis*), ~1,200 proteins
- `bg_rat.rds` — Rat (*Rattus norvegicus*), ~8,200 proteins

Files from a previous v0.7 installation can be reused without rebuilding.

2.3 Launch

```
shiny::runApp("app.R")
```

The app opens in your default browser. A built-in Trastuzumab heavy chain + light chain FASTA is pre-loaded so you can explore the workflow immediately.

3 Workflow Step 1: Input and Setup (Tab 1)

3.1 Load the ADC Sequence

Two input modes are available:

- **Upload File** — upload a `.fasta`, `.fa`, or `.txt` file containing one or more `>header / sequence` entries. Each entry is treated as a separate chain.
- **Paste Sequence** — paste FASTA text directly. The demo Trastuzumab sequence is pre-loaded; replace it with your own ADC.

A chain preview table appears after parsing, confirming the number of chains, their headers, lengths (AA), and first 60 residues.

ADC Name: enter a short identifier (e.g. `Trastuzumab-DXd`). This is used in all output filenames and as the `ADCName` column in exported transition lists.

3.2 Digest Settings

Setting	Recommended for tryptic mapping	Notes
Primary enzyme	Trypsin	Cleaves after K/R, not before P
Second enzyme	None	Set to Lys-C for Trypsin+Lys-C
Missed cleavages	0 or 1	0 for standard mapping; 1 to increase coverage
Peptide length filter	6–30 AA	Fixed; cannot be changed

3.3 Background Proteome

Select **Human (Homo sapiens)** for human ADC PK studies. For preclinical species comparison, select Cynomolgus Monkey or Rat. A custom FASTA upload is available for other species.

The **Use missed cleavage level** dropdown for the background sets how many missed cleavages were allowed when the background was digested. Setting it to match the digest missed cleavage level gives the most conservative (strictest) uniqueness call.

3.4 Run Digest

Click **Run Digest & Uniqueness Check** (at the bottom of the FASTA Input card). The app digests all chains, applies modifications, checks uniqueness against the selected background, and navigates to Tab 3. A notification reports:

- Total peptide variants generated
 - Number unique to the ADC (not in background)
-

4 Workflow Step 2: Modifications (Tab 2)

4.1 Fixed and Variable Modifications

Carbamidomethylation (+57.021 Da on C) is always applied. Select variable modifications to include in the digest:

- **Oxidation (M)** — standard variable mod for methionine sulfoxide
- **Drug-Linker Payload** — select from the ADCDB dropdown (MMAE, DM1, DXd, SN-38, etc.) or enter a custom mass shift. Applied to cysteine (C) residues at conjugation sites.

4.2 Linker Biotransformation Modifications

For ADC linker stability studies, enable one or more biotransformation mods:

Modification	Mass shift	Scientific context
Maleimide ring hydrolysis	+18.011 Da on C	Succinimide-thioether hydrolysis in plasma
Succinimide ring-opening	+18.011 Da on C	Ring-opened adduct at pH 7.4
Thioether to sulfoxide	+15.995 Da on C	Oxidative stress
Disulfide loss	-31.990 Da on C	Retro-Michael elimination
Conjugation-site deamidation	+0.984 Da on N	Asn deamidation near conjugation site

These appear as separate peptide variants in the Tab 3 table and in the transition list.

4.3 DAR Distribution Settings

Set the conjugation chemistry and maximum DAR level. The app generates separate precursor entries for each DAR species (DAR0, DAR1, ..., DAR_n), each with the appropriate number of drug-linker mass units added to conjugated peptide(s).

5 Workflow Step 3: Peptide Results (Tab 3)

5.1 Interpreting the Peptide Table

The table columns are:

Column	Description
Chain	FASTA header of the source chain
Peptide Sequence	Unmodified sequence
Modified Sequence	Sequence with brackets, e.g. CASC[CAM]K
Start / End	1-based positions in the chain
Length	Number of amino acids
Modifications	Applied PTMs, e.g. CAM@C3; DXd@C3
Mass (Da)	Monoisotopic mass
Unique to ADC	Navy badge = unique; grey badge = found in background

5.2 Filtering

Use the dropdown filters above the table to restrict to:

- A specific chain (Heavy Chain, Light Chain)
- Unique peptides only (preferred for bioanalytical method development)
- Modified peptides only

5.3 Sequence Coverage Map

The coverage map card appears below the table after the digest. Use it to:

- Identify gaps (regions not covered by any peptide in the 6–30 AA window)
- Colour by uniqueness to see which regions have ADC-specific peptides
- Increase missed cleavages in Tab 1 to close coverage gaps

Download the map as a 300 dpi PNG for use in reports.

5.4 Cross-Species Co-Uniqueness

For multi-species PK/TK studies, use the **Co-uniqueness Check** panel to identify peptides unique across all selected species simultaneously. This prevents selection of surrogate peptides that cross-react with the preclinical species matrix proteome.

6 Workflow Step 4: Transition List (Tab 4)

6.1 Instrument Selection

Select the target instrument platform from the dropdown. The app adjusts the collision energy formula and CSV column layout automatically:

Platform	CE type	Use case
Skyline (generic)	Sciex-style	Import into Skyline for further optimisation
Thermo (Orbitrap/TSQ)	Linear, charge-dependent	TSQ Altis, TSQ Quantiva
SCIEX (QTRAP/TripleTOF)	Empirical	QTRAP 6500+, TripleTOF 6600
Bruker (timsTOF)	TIMS-adjusted	timsTOF Pro, timsTOF SCP
Agilent (QQQ/QTOF)	Agilent empirical	6495, 6560
Waters (Xevo/Synapt)	Waters empirical	Xevo TQ-S, Synapt G2-Si

6.2 DAR Level Selection

Select one or more DAR levels (DAR0, DAR1, ..., DARn) to include. Each level generates separate rows in the transition list with the corresponding precursor m/z.

6.3 Ion Parameters

Fragment ions are b2–b(n-1) and y2–y(n-1), singly charged. Precursor charge states 2+, 3+, and 4+ are generated. Use **Top N ions per precursor** to limit the list to the strongest predicted ions for a given platform.

6.4 Download

For Skyline-based method development:

1. Download **Skyline CSV (Unique Only)**.
2. In Skyline: **File** → **Import** → **Transition List** → **Column headers match** and select the CSV.

For direct instrument method setup, download the **Instrument CSV** for your platform or the **Excel Summary** for a complete multi-sheet workbook.

7 Workflow Step 5: Heavy Labelling (Tab 5)

Heavy stable isotope labelling (SIL) peptides serve as internal standards (SIL-IS) for absolute quantification by LC-MS/MS.

7.1 Typical SILAC-style experiment

1. Select **¹³C6 ¹⁵N2 Lys** (+8.014 Da) for tryptic peptides ending in K, and/or **¹³C6 ¹⁵N4 Arg** (+10.008 Da) for peptides ending in R.
 2. Select charge states 2+ and 3+.
 3. Download **Heavy Peptides Excel**.
 4. Use the Light/Heavy Pair Summary sheet to verify the expected Delta m/z (e.g. 4.007 Th for a doubly charged peptide labelled with ¹³C6 ¹⁵N2 Lys).
-

8 Workflow Step 6: MS/MS Search Confirmation (Tab 6)

After acquiring LC-MS/MS data on an ADC tryptic digest, use Tab 6 to confirm peptide identifications against the theoretical list from Tab 3.

8.1 Recommended workflow

1. Convert raw files to mzML with MSConvert:

```
msconvert sample.raw --mzML --filter "peakPicking true 1-"
```
2. Upload the mzML file in Tab 6.
3. Ensure the correct engine is detected (MS Amanda badge = green).
4. Click **Run Search**. The app generates `settings.xml` from your Tab 1/2 parameters and launches MS Amanda.
5. After the search, review the **Detected vs Theoretical** sub-tab and adjust the score threshold slider to balance sensitivity and specificity.
6. Check the target-decoy FDR estimate. For quantitative bioanalytical methods, aim for FDR $\leq 1\%$.
7. Download the filtered PSM table and coverage summary.

8.2 Pre-computed results

If you have existing MS Amanda output (.mzid) or Tide/Crux output (.pepxml), upload them directly. The search step is skipped and results are cross-referenced immediately.

9 Workflow Step 7: AI Assistant (Tab 7)

The AI Assistant uses the Anthropic Claude API to provide contextual guidance.

9.1 Example questions

- “Which of my unique peptides would be the best MRM surrogate for the heavy chain?”
- “Explain why CASIQKFGR is commonly used as an IgG1 reference peptide.”
- “What does a coverage gap between residues 220–240 of my heavy chain suggest?”
- “How do I optimise collision energy for a 3+ precursor on a SCIEX QTRAP 6500+?”

9.2 Setup

Enter your Anthropic API key and click **Save to .Renviron** (one-time). Enable **Include digest context** to attach the current ADC name, enzyme, and peptide count automatically. The model list is populated from your account’s available models.

10 Interpreting Results: Common Scenarios

10.1 Scenario A: No unique peptides found

Possible causes:

- The ADC is based on a human antibody scaffold and shares most tryptic peptides with the human proteome background.
- The drug-linker payload modification was not selected, so conjugated peptides are being compared without their payload mass.

Suggested actions:

- Enable the drug-linker payload modification in Tab 2 and re-run the digest. Payload-bearing peptides are typically unique because the modification mass is not present in the background.
- Reduce the background missed cleavage level (Tab 1 Background section) to allow a more permissive uniqueness call.
- Try a different enzyme (e.g. Lys-C or Asp-N) to generate peptides covering different sequence regions.

10.2 Scenario B: Poor sequence coverage

Possible causes:

- The default missed cleavage level (0) generates short peptides that fall outside the 6–30 AA window.
- The enzyme selected does not have cleavage sites in certain regions.

Suggested actions:

- Increase missed cleavages to 1 or 2 in Tab 1 and re-run.
- Try a second enzyme (e.g. Glu-C) to generate complementary peptides.

10.3 Scenario C: DAR heterogeneity in the transition list

When DAR modelling is enabled, you will see multiple precursor m/z entries for the same peptide sequence at different DAR levels. This is expected and correct:

- **DAR0** entries correspond to unconjugated (naked antibody) peptides.
- **DAR1, DAR2, ...** entries correspond to singly, doubly, ... conjugated species.

For quantification of DAR distribution by LC-MS/MS, monitor at least DAR0 and the dominant DAR species (typically DAR2 for cysteine-conjugated ADCs, DAR4 for higher loading).

11 Running Tests

Before deploying or distributing, verify the mass engine is working correctly:

```
source("R/digest.R")
source("R/modifications.R")
source("R/transitions.R")
source("R/isotopes.R")
source("R/dar.R")

# 79 unit tests
source("tests/test_masses.R")

# 10 mass accuracy benchmarks (sub-mDa vs IgG1 reference peptides)
source("tests/benchmark_mass_accuracy.R")
```

Expected output: 79 / 79 PASSED and 10 / 10 PASSED.

12 Citation

Wase, N. (2026). ADC Peptide Mapper (Version 0.8) [Software].
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See `CITATION.cff` in the project root for the full reference list including all scientific citations for the mass accuracy models, FDR method, and linker biotransformation references.

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For research use only. All masses are monoisotopic.